

Problem 8.10. The Japanese game *go-moku* is played by two players, “X” and “O” on a 19×19 grid. Players take turns placing markers, and the first player to achieve five of her markers consecutively in a row, column, or diagonal is the winner. Consider this game generalized to an $n \times n$ board. Let

$GM = \{\langle B \rangle \mid B \text{ is a position in generalized go-moku, where player “X” has a winning strategy}\}.$

By a *position* we mean a board with markers placed on it, such as may occur in the middle of a play of the game, together with an indication of which player moves next. Show that $GM \in PSPACE$.

Proof. To show that $GM \in PSPACE$, we give a polynomial space algorithm M that decides GM .

$M =$ “On input $\langle B \rangle$, where B is a position in generalized go-moku:

1. If $n < 5$, then *reject*.
2. If player “O” has won, then *reject*. If player “X” has won, then *accept*.
3. If player “X” moves next, then:
 4. Repeat for each $i = 1$ to n .
 5. Repeat for each $j = 1$ to n .
 6. If no marker is placed on location (i, j) , then:
 7. Place marker X on (i, j) , and mark player “O” moves next.
 8. Recursively call M on $\langle B \rangle$.
 9. If M accepts, then *accept*. Otherwise, remove marker X from location (i, j) .
10. *Reject*.
11. If player “O” moves next, then:
 12. Repeat for each $i = 1$ to n .
 13. Repeat for each $j = 1$ to n .
 14. If no marker is placed on location (i, j) , then:
 15. Place marker O on (i, j) , and mark player “X” moves next.
 16. Recursively call M on $\langle B \rangle$.
 17. If M rejects, then *reject*. Otherwise, remove marker X from location (i, j) .
18. *Accept.*”

□